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Wheel and mobility including the same

Abstract

A wheel includes a deformable part reversibly deformable in shape by an external force, wherein the deformable part includes an inner region having a shape extending in a peripheral direction, an outer region having a shape extending in the peripheral direction and spaced apart outward from the inner region in a radial direction, and a spoke region provided between the inner region and the outer region and having a first side connected to the inner region and a second side connected to the outer region. The spoke region includes a first spoke region having a section protruding in a first of two peripheral directions and a second spoke region disposed at a first side of the first spoke region based on a second of the two peripheral directions opposite the first peripheral direction, wherein the second spoke region has a section protruding in the second peripheral direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of Korean Patent Application No. 10-2022-0059824, filed on May 16, 2022, which application is hereby incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a wheel and a mobility including the same.

BACKGROUND

(3) As a demand for a mobility including a delivery robot other than traditional vehicles in the related art increases, studies are also being actively conducted on a wheel mounted on the mobility. In particular, because the mobility is often required to be movable on various terrains including stairs or obstacles as well as flat ground, there is an increasing demand for a wheel capable of moving in various types of environments.

(4) In the related art, there have been disclosed an airless type wheel having a structure into which no air is injected so as to be movable in various types of environments and an active wheel having a structure capable of actively controlling a shape thereof suitably for the terrain.

(5) However, the airless type wheel is not greatly deformed in shape because of the nature of a physically hard tire, which makes it difficult for the airless type wheel to move on stairs. The active wheel requires devices for controlling the shape of the wheel, which excessively complicates the structure of the wheel.

(6) Therefore, there is a need to develop a new type of wheel having a simple structure while being capable of smoothly moving even on a complicated terrain such as stairs.

SUMMARY

(7) The present disclosure relates to a wheel and a mobility including the same. Particular embodiments relate to a wheel having a structure capable of stably moving in various situations and a mobility including the same.

(8) Embodiments of the present disclosure provide a new type of wheel having a simple structure while being capable of smoothly moving even on various terrains such as stairs.

(9) An exemplary embodiment of the present disclosure provides a wheel including a deformable part configured to be reversibly deformed in shape by an external force, in which the deformable part includes an inner region having a shape extending in a peripheral direction P, an outer region having a shape extending in the peripheral direction and spaced apart outward from the inner region in a radial direction R, and a spoke region provided between the inner region and the outer region and having one side connected to the inner region and the other side connected to the outer region. The spoke region includes a first spoke region having a section protruding in a first peripheral direction P1 of two directions of the peripheral direction P and a second spoke region disposed at one side of the first spoke region based on a second peripheral direction P2 opposite to the first peripheral direction P1 of the two directions of the peripheral direction P and having a section protruding in the second peripheral direction P2.

(10) The spoke region may be provided in plural, and the plurality of spoke regions may be disposed in the peripheral direction P.

(11) The first spoke region may be longer than the second spoke region.

- (12) The first spoke region may be thicker than the second spoke region.
- (13) An interval A1 between the outer region and a section of the first spoke region, which protrudes by the largest amount in the first peripheral direction P1, may be smaller than an interval A2 between the inner region and a section of the first spoke region that protrudes by the largest amount in the first peripheral direction P1.
- (14) An interval B1 between the outer region and a section of the second spoke region, which protrudes by the largest amount in the second peripheral direction P2, may be smaller than an interval B2 between the inner region and a section of the second spoke region that protrudes by the largest amount in the second peripheral direction P2.
- (15) An interval A1 between the outer region and a section of the first spoke region, which protrudes by the largest amount in the first peripheral direction P1, may be larger than an interval B1 between the outer region and a section of the second spoke region that protrudes by the largest amount in the second peripheral direction P2.
- (16) An interval D1 between a region in which the first spoke region and the outer region meet and a region in which the second spoke region and the outer region meet may be larger than an interval D2 between a region in which the first spoke region and the inner region meet and a region in which the second spoke region and the inner region meet.
- (17) A thickness of the first spoke region at a portion where the first spoke region is connected to the inner region may be larger than a thickness of the first spoke region at a portion where the first spoke region is connected to the outer region.
- (18) A thickness of the second spoke region at a portion where the second spoke region is connected to the inner region may be larger than a thickness of the second spoke region at a portion where the second spoke region is connected to the outer region.
- (19) A portion where the first spoke region is connected to the outer region may be directed away in the peripheral direction P from a portion where the first spoke region is connected to the inner region.
- (20) A portion where the second spoke region is connected to the outer region may be directed away in the peripheral direction P from a portion where the second spoke region is connected to the inner region.
- (21) A portion where the first spoke region is connected to the outer region may be directed away in the second peripheral direction P2 from a portion where the first spoke region is connected to the inner region.
- (22) A portion where the second spoke region is connected to the outer region may be directed away in the second peripheral direction P2 from a portion where the second spoke region is connected to the inner region.
- (23) The plurality of spoke regions may be disposed at intervals that allow the first spoke region, which is provided in one spoke region among the plurality of spoke regions, and the second spoke region of the other spoke region spaced apart from the one spoke region in the first peripheral direction P1 to come into close contact with each other when an external force of a predetermined magnitude or higher is applied to the outer region in the radial direction R.
- (24) The wheel may further include a core part provided inside the inner region in the radial direction R and having one side fixedly coupled to the inner region and a connection part having one side connected to the core part and the other side connected to the outer region.
- (25) Another exemplary embodiment of the present disclosure provides a wheel including a deformable part configured to be reversibly deformed in shape by an external force, in which the deformable part includes an inner region having a shape extending in a peripheral direction P, an outer region having a shape extending in the peripheral direction P and spaced apart outward from the inner region in a radial direction R, and spoke regions provided between the inner region and the outer region and each having one side connected to the inner region and the other side connected to the outer region, in which the spoke regions are spaced apart from one another in the

peripheral direction P. The spoke regions each include a first spoke portion extending outward from the inner region in the radial direction R, a second spoke portion extending from the first spoke portion in a first peripheral direction P1 of two directions of the peripheral direction P, and a third spoke portion extending outward from the second spoke portion in the radial direction R and connected to the outer region.

(26) The first spoke portion may extend outward from the inner region in the radial direction R and extend to be directed in a second peripheral direction P2 opposite to the first peripheral direction P1 of the peripheral direction P, and the third spoke portion may extend outward from the second spoke portion in the radial direction R and extend to be directed in the second peripheral direction P2.

(27) A region Z23 in which the second spoke portion and the third spoke portion of one spoke region provided in the deformable part meet may be spaced apart outward, in the radial direction R, from a region Z12 in which the first spoke portion and the second spoke portion meet in another spoke region of the two spoke regions that is spaced apart from one spoke region in the first peripheral direction P1 of the peripheral direction P.

(28) Still another exemplary embodiment of the present disclosure provides a mobility including a wheel and a frame member to which the wheel is coupled, in which the wheel includes a deformable part configured to be reversibly deformed in shape by an external force, in which the deformable part includes an inner region having a shape extending in a peripheral direction P, an outer region having a shape extending in the peripheral direction P and spaced apart outward from the inner region in a radial direction R, and a spoke region provided between the inner region and the outer region and having one side connected to the inner region and the other side connected to the outer region. The spoke region includes a first spoke region having a section protruding in a first peripheral direction P1 of two directions of the peripheral direction P and a second spoke region disposed at one side of the first spoke region based on a second peripheral direction P2 opposite to the first peripheral direction P1 of the two directions of the peripheral direction P and having a section protruding in the second peripheral direction P2.

(29) According to embodiments of the present disclosure, it is possible to provide a new type of wheel having a simple structure while being capable of smoothly moving even on various terrains such as stairs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view illustrating a wheel according to an embodiment of the present disclosure.

(2) FIG. 2 is a view illustrating a deformable part of the wheel according to an embodiment of the present disclosure.

(3) FIG. 3 is an enlarged view of a spoke region of the deformable part of the wheel according to an embodiment of the present disclosure.

(4) FIG. 4 is a view illustrating a shape of the deformable part when the wheel according to an embodiment of the present disclosure moves on the flat ground.

(5) FIG. 5 is an enlarged view of the spoke region illustrated in FIG. 4.

(6) FIG. 6 is a view illustrating a shape of the deformable part when the wheel according to an embodiment of the present disclosure moves up the stairs.

(7) FIG. 7 is an enlarged view of the spoke region illustrated in FIG. 6.

(8) FIG. 8 is a view illustrating a deformable part of a wheel according to another embodiment of the present disclosure.

(9) FIG. 9 is a view illustrating a shape of the deformable part when the wheel according to another embodiment of the present disclosure moves on the flat ground.

(10) Hereinafter, a wheel and a mobility according to embodiments of the present disclosure will be described with reference to the drawings.

(11) Wheel

(12) FIG. 1 is a view illustrating a wheel according to an embodiment of the present disclosure, and FIG. 2 is a view illustrating a deformable part of the wheel according to an embodiment of the present disclosure. FIG. 3 is an enlarged view of a spoke region of the deformable part of the wheel according to an embodiment of the present disclosure.

(13) A wheel **100** according to embodiments of the present disclosure may ensure stable movement performance by generating a resistive force against an external force when the wheel **100** moves on the flat ground. Further, the wheel **100** may improve ride quality as a trajectory of a center of the wheel has an inclined shape (slope) with a predetermined gradient without having a level difference when the wheel moves in a section having stepped portions such as stairs.

(14) More specifically, referring to FIGS. 1 to 3, the wheel **100** according to embodiments of the present disclosure may include a deformable part **110** configured to be reversibly deformed in shape by an external force. That is, when the external force is applied to the deformable part **110**, the deformable part **110** may be deformed in shape. When the external force is eliminated, the deformable part **110** may return to an original state. Therefore, the deformable part **110** may include a material having predetermined elasticity. For example, the deformable part **110** may include a rubber material. For example, the deformable part **110** may include styrene butadiene rubber (SBR) or may be made of SBR. However, the material of the deformable part **110** is not limited as long as the material has predetermined elasticity. Meanwhile, as illustrated in FIGS. 1 to 3, peripheries of the wheel **100** and the deformable part **110** according to embodiments of the present disclosure may each have an approximately circular shape in a state in which no external force is applied.

(15) The components of the deformable part **110** may be integrated. However, according to embodiments of the present disclosure, the deformable part **110** may be divided into a plurality of regions.

(16) More specifically, the deformable part **110** may include an inner region **112** having a shape extending in a peripheral direction P. For example, the inner region **112** may have a ring shape having a predetermined radius.

(17) In addition, the deformable part **110** may further include an outer region **114** having a shape extending in the peripheral direction P and spaced apart outward from the inner region **112** in a radial direction R of the wheel **100**. The outer region **114** may have a ring shape having a larger radius than the inner region **112**. The outer region **114** is a region that first receives an external force from a ground surface, among the regions of the deformable part **110**, during a process in which the wheel **100** moves. The outer region **114** may primarily support a load. More particularly, an approximately concentric relationship may be defined between the inner region **112** and the outer region **114**.

(18) In addition, the deformable part **110** may further include a spoke region **116** provided between the inner region **112** and the outer region **114** in the radial direction R and having one side connected to the inner region **112** and the other side connected to the outer region **114**. Referring to the above-mentioned description, the inner region **112**, the outer region **114**, and the spoke region **116** may be integrated with one another.

(19) Referring to FIGS. 1 and 2, the spoke region **116** may be provided in plural, and the plurality of spoke regions **116** may be disposed in the peripheral direction P of the wheel **100**. More particularly, the plurality of spoke regions **116** may be provided at equal intervals in the peripheral direction P of the wheel **100**.

(20) The plurality of spoke regions **116** may each be divided into a plurality of regions. More specifically, the spoke region **116** may include a first spoke region **116a** having a section protruding

in a first peripheral direction P1 of the two directions of the peripheral direction P, and a second spoke region **116b** disposed at one side of the first spoke region **116a** based on a second peripheral direction P2 opposite to the first peripheral direction P1 of the two directions of the peripheral direction P and having a section protruding in the second peripheral direction P2. For example, as illustrated in FIGS. 1 to 3, a predetermined space may be defined between the first spoke region **116a** and the second spoke region **116b** in one spoke region **116**, and a predetermined space may also be defined between the two adjacent spoke regions **116**. As described below, the predetermined spaces may serve as spaces in which the spoke regions **116** are deformed when the external force is applied to the wheel **100**.

(21) According to an exemplary embodiment of the present disclosure, the plurality of spoke regions **116** of the wheel **100** may have the same shape or have the shapes corresponding to one another. In this case, the configuration in which the spoke regions **116** have the shapes corresponding to one another may be understood as including a case in which the spoke regions **116** have the same shape and a case in which the spoke regions **116** have shapes similar to one another to the extent that the spoke regions **116** have substantially the same function. However, the first and second spoke regions **116a** and **116b** provided in one spoke region **116** may be different in geometric shape. Hereinafter, the shapes of the first and second spoke regions **116a** and **116b** will be described in detail with reference to the drawings.

(22) FIG. 4 is a view illustrating a shape of the deformable part when the wheel according to an embodiment of the present disclosure moves on the flat ground, and FIG. 5 is an enlarged view of the spoke region illustrated in FIG. 4. FIG. 6 is a view illustrating a shape of the deformable part when the wheel according to an embodiment of the present disclosure moves up the stairs, and FIG. 7 is an enlarged view of the spoke region illustrated in FIG. 6.

(23) Referring to FIGS. 4 to 7, when an external force is applied to the outer region **114**, the spoke regions **116** provided between the inner region **112** and the outer region **114** are deformed in shape while a distance between the inner region **112** and the outer region **114** decreases. More specifically, the spoke regions **116** between the inner region **112** and the outer region **114** are pressed, such that the first and second spoke regions **116a** and **116b** may have crushed shapes. Meanwhile, examples in which the external force is applied to the outer region **114** may include a case in which the wheel including the deformable part **110** moves on a flat ground surface as illustrated in FIGS. 4 and 5, a case in which the wheel including the deformable part **110** moves up the stairs as illustrated in FIGS. 6 and 7, and a case in which the ground surface or stair presses the outer region **114**.

(24) In this case, as described above, the first spoke region **116a** has the section protruding in the first peripheral direction P1, and the second spoke region **116b** has the section protruding in the second peripheral direction P2. Therefore, according to embodiments of the present disclosure, the two adjacent spoke regions **116** may come into close contact with each other when the external force is applied to the deformable part **110** through the outer region **114**.

(25) More specifically, referring to FIGS. 4 to 7, the plurality of spoke regions **116** may be disposed at intervals that allow the first spoke region **116a** of one spoke region **116-1** of the plurality of spoke regions **116** and the second spoke region **116b** of the other spoke region **116-2**, which is spaced apart from one spoke region **116-1** in the first peripheral direction P1, to come into close contact with each other when an external force of a predetermined magnitude or higher is applied to the outer region **114** in the radial direction R.

(26) That is, referring to FIGS. 4 to 7, when an external force of a predetermined magnitude or higher is applied to the outer region **114** of the wheel **100** according to embodiments of the present disclosure, the two adjacent spoke regions **116** come into close contact with each other, such that the two spoke regions **116** may support each other, and the two spoke regions **116** may be prevented from being additionally deformed. Therefore, according to embodiments of the present disclosure, because the two adjacent spoke regions **116** support each other, a degree of deformation

in the shape of the deformable part **110** may be reduced. Further, because a displacement of the mobility equipped with the wheel **100** may be minimized during the process in which the wheel **100** moves, the movement performance of the wheel **100** may be ensured.

(27) In FIG. **4**, the dotted line indicates a part of an imaginary circle having a center corresponding to a rotation center of the deformable part **110** and including a region in which the two adjacent spoke regions **116** come into close contact with each other while the wheel **100** moves on the ground surface. In FIG. **6**, the dotted line indicates a part of an imaginary circle having a center corresponding to the rotation center of the deformable part **110** and including a region in which the two adjacent spoke regions **116** come into close contact with each other while the wheel **100** moves on the stairs. That is, according to embodiments of the present disclosure, the outer region **114** may primarily support a load of the wheel **100** and a load of the mobility equipped with the wheel **100** during the process in which the wheel **100** moves. The region in which the two adjacent spoke regions **116** come into close contact with each other and support each other may secondarily support the load of the wheel **100** and the load of the mobility. For example, the region in which the two spoke regions **116** come into close contact with each other and support each other may be a part of an imaginary load support member defined along the dotted line illustrated in FIGS. **4** and **6**. However, the two spoke regions **116**, which support each other while coming into close contact with each other, may be continuously changed over time because the deformable part **110** rotates during the process in which the wheel **100** moves.

(28) More particularly, according to embodiments of the present disclosure, when the mobility equipped with the wheel **100** moves forward, the wheel **100** may rotate in the second peripheral direction P2. In addition, as illustrated in FIGS. **1** to **3**, in one spoke region **116**, the first spoke region **116a** may be longer than the second spoke region **116b**, and the first spoke region **116a** may be thicker than the second spoke region **116b**.

(29) As described above, when the wheel **100** rotates in the second peripheral direction P2, the first spoke region **116a** of one spoke region **116** faces the ground surface before the second spoke region **116b** faces the ground surface. Therefore, when the wheel **100** rotates in the second peripheral direction P2, the first spoke region **116a** is deformed first.

(30) Therefore, to ensure the movement performance of the wheel **100** according to embodiments of the present disclosure, the deformation amount of the first spoke region **116a**, which is deformed first before the second spoke region **116b** is deformed during the process in which the wheel **100** moves, needs to be limited within a predetermined range. Therefore, an elastic modulus of the first spoke region **116a** needs to be larger than an elastic modulus of the second spoke region **116b**. In this case, the elastic modulus may be understood as the concept indicating a degree of deformation in shape when an external force is applied to the first spoke region **116a** or the second spoke region **116b**.

(31) In this case, as described above, according to embodiments of the present disclosure, because the first spoke region **116a** is thicker than the second spoke region **116b**, the first spoke region **116a** may have a relatively large elastic modulus. Therefore, it is possible to minimize a degree to which the first spoke region **116a** is deformed in shape by the external force during the process in which the wheel **100** moves. As a result, when the external force is applied, the first spoke region **116a** may more securely support the wheel **100** and the mobility equipped with the wheel **100**, and the movement performance of the wheel **100** may also be improved.

(32) Meanwhile, as described above, in one spoke region **116**, the first spoke region **116a** may be longer than the second spoke region **116b**. In this case, it is possible to maximize a degree to which the first spoke region **116a** protrudes in the first peripheral direction P1 when an external force is applied to the first spoke region **116a** during the process in which the wheel **100** moves. Therefore, the first spoke region **116a** may quickly meet the adjacent second spoke region **116b**, such that the process in which the first and second spoke regions **116a** and **116b** support each other while coming into close contact with each other may be quickly performed.

(33) Meanwhile, referring to FIGS. 1 to 3, in each of the spoke regions **116**, an interval A1 between the outer region **114** and a section of the first spoke region **116a**, which protrudes by the largest amount in the first peripheral direction P1, may be smaller than an interval A2 between the inner region **112** and a section of the first spoke region **116a** that protrudes by the largest amount in the first peripheral direction P1. In addition, in each of the spoke regions **116**, an interval B1 between the outer region **114** and a section of the second spoke region **116b**, which protrudes by the largest amount in the second peripheral direction P2, may be smaller than an interval B2 between the inner region **112** and a section of the second spoke region **116b** that protrudes by the largest amount in the second peripheral direction P2.

(34) Referring to FIGS. 4 to 7, when the two adjacent spoke regions **116-1** and **116-2** come into close contact with each other as an external force is applied to the wheel **100**, a section of the first spoke region **116a** of one spoke region **116-1** of the two spoke regions **116-1** and **116-2** coming into close contact with each other, which protrudes by the largest amount in the first peripheral direction P1, comes into close contact with a section of the second spoke region **116b** of the other spoke region **116-2** that protrudes by the largest amount in the second peripheral direction P2.

(35) In this case, as described above, A1 is smaller than A2, and B1 is smaller than B2, such that the region in which the first spoke region **116a** of one spoke region **116-1** and the second spoke region **116b** of the other spoke region **116-2** come into close contact with each other may be positioned relatively toward the outside in the radial direction R of the wheel **100**. Therefore, a degree of deformation of the deformable part **110** may be minimized. This configuration may correspond to the configuration in which the dotted line illustrated in FIGS. 4 and 6 is closer to the outer region **114** than is the inner region **112**.

(36) Meanwhile, referring to FIG. 3, the interval A1 between the outer region **114** and the section of the first spoke region **116a**, which protrudes by the largest amount in the first peripheral direction P1, may be larger than the interval B1 between the outer region **114** and the section of the second spoke region **116b** that protrudes by the largest amount in the second peripheral direction P2.

(37) Therefore, as illustrated in FIGS. 4 to 7, when the first spoke region **116a** of one spoke region **116-1** of the two adjacent spoke regions comes into close contact with the second spoke region **116b** of the other spoke region **116-2** as the spoke regions **116** are deformed by the external force applied to the wheel **100**, the section of the second spoke region **116b** of the other spoke region **116-2**, which protrudes by the largest amount in the second peripheral direction P2, is positioned outward in the radial direction R from the section of the first spoke region **116a** of one spoke region **116-1** that protrudes by the largest amount in the first peripheral direction P1. In this case, the first spoke region **116a** of one spoke region **116-1** may press, outward in the radial direction R, the second spoke region **116b** of the other spoke region **116-2**. The second spoke region **116b** of the other spoke region **116-2** may press, inward in the radial direction R, the first spoke region **116a** of one spoke region **116-1**.

(38) In addition, according to embodiments of the present disclosure, as illustrated in FIG. 3, in one spoke region **116**, an interval D1 between a region in which the first spoke region **116a** and the outer region **114** meet and a region in which the second spoke region **116b** and the outer region **114** meet may be larger than an interval D2 between a region in which the first spoke region **116a** and the inner region **112** meet and a region in which the second spoke region **116b** and the inner region **112** meet. Therefore, as illustrated in FIG. 3, in one spoke region **116**, a vacant space defined by the first and second spoke regions **116a** and **116b** may have an approximately water droplet shape. Therefore, when the spoke region **116** is deformed by an external force, a space, which is adjacent to the outer region **114** among spaces between the inner region **112** and the outer region **114**, is sufficiently provided to allow the spoke region **116** to be deformed.

(39) Referring to FIG. 3, a thickness of the first spoke region **116a** at a portion where the first spoke region **116a** is connected to the inner region **112** may be larger than a thickness of the first

spoke region **116a** at a portion where the first spoke region **116a** is connected to the outer region **114**. Further, a thickness of the second spoke region **116b** at a portion where the second spoke region **116b** is connected to the inner region **112** may be larger than a thickness of the second spoke region **116b** at a portion where the second spoke region **116b** is connected to the outer region **114**. Therefore, when the spoke region **116** is deformed by an external force, the inner region **112** may securely support the first and second spoke regions **116a** and **116b**.

(40) Meanwhile, according to embodiments of the present disclosure, the portion where the first spoke region **116a** is connected to the outer region **114** may be directed away in the peripheral direction P from the portion where the first spoke region **116a** is connected to the inner region **112**. The portion where the second spoke region **116b** is connected to the outer region **114** may be directed away in the peripheral direction P from the portion where the second spoke region **116b** is connected to the inner region **112**.

(41) More particularly, as illustrated in FIGS. 1 to 3, the portion where the first spoke region **116a** is connected to the outer region **114** may be directed away in the second peripheral direction P2 from the portion where the first spoke region **116a** is connected to the inner region **112**. The portion where the second spoke region **116b** is connected to the outer region **114** may be directed away in the second peripheral direction P2 from the portion where the second spoke region **116b** is connected to the inner region **112**. Because the first and second spoke regions **116a** and **116b** are directed in the peripheral direction P as described above, the first and second spoke regions **116a** and **116b** of the spoke region **116** may always be deformed in a predetermined direction during the process in which the wheel **100** moves, thereby ensuring that the two adjacent spoke regions **116** come into close contact with each other.

(42) Meanwhile, as illustrated in FIG. 1, the wheel **100** according to embodiments of the present disclosure may further include a core part **120** disposed inside the inner region **112** in the radial direction R and having one side fixedly coupled to the inner region **112**. The core part **120** may be configured to support the deformable part **110**. For example, as illustrated in FIG. 1, the inner region **112** may be inserted into the core part **120**.

(43) In addition, the wheel **100** may further include connection parts **130** each having one side connected to the core part **120** and the other side connected to the outer region **114**. For example, the connection part **130** may have a rod shape. The connection part **130** may be provided in plural, and the plurality of connection parts **130** may be disposed in the peripheral direction P.

(44) Hereinafter, a situation in which the deformable part **110** is deformed when the wheel according to embodiments of the present disclosure moves on a flat ground surface and a situation in which the deformable part **110** is deformed when the wheel moves on the stairs having a stepped shape difference will be described with reference to FIGS. 4 to 7.

(45) As illustrated in FIGS. 4 and 5, when the wheel moves on the flat ground surface, the configuration in which running resistance of the wheel against the ground surface is low is advantageous in terms of efficiency. Therefore, in this case, the deformable part **110** needs to be less deformed. To this end, as described above, the two adjacent spoke regions **116** come into close contact with each other in the region adjacent to the outer region **114**. In this case, the two adjacent spoke regions **116** may come into close contact with each other in a relatively large number of regions in order to disperse the load of the wheel and the load of the mobility. For example, FIG. 4 illustrates that the two spoke regions **116** come into close contact with each other at four points.

(46) In contrast, as illustrated in FIGS. 6 and 7, when the wheel moves on the stairs, the deformable part **110** needs to be greatly deformed correspondingly to the stepped shape of the stairs. Therefore, in this case, the two adjacent spoke regions **116** may come into close contact with each other in a smaller number of regions than when the wheel moves on the flat ground surface. For example, FIG. 6 illustrates that the two spoke regions **116** come into close contact with each other at two points. When the two adjacent spoke regions **116** come into close contact with each other in a small number of regions, the load of the wheel and the load of the mobility, which are applied to the

region in which the two spoke regions **116** come into close contact with each other, increase. Therefore, in the region in which the two spoke regions **116** come into close contact with each other, the first spoke region **116a** or the second spoke region **116b** may be relatively greatly deformed, and thus the shape of the deformable part **110** may be deformed to conform to the shape of the stairs.

(47) FIG. **8** is a view illustrating a deformable part of a wheel according to another embodiment of the present disclosure, and FIG. **9** is a view illustrating a shape of the deformable part when the wheel according to another embodiment of the present disclosure moves on the flat ground.

(48) The above-mentioned contents related to the wheel according to the embodiment of the present disclosure may be equally applied to the wheel according to another embodiment of the present disclosure. However, another embodiment of the present disclosure differs from the above-mentioned embodiment of the present disclosure in terms of a shape of the spoke region **116** of the deformable part **110**.

(49) As illustrated in FIG. **8**, another embodiment of the present disclosure differs from the above-mentioned embodiment of the present disclosure in that all of the plurality of spoke regions **116** provided in the deformable part **110** of the wheel and disposed to be spaced apart from one another in the peripheral direction P have substantially the same shape.

(50) More specifically, as illustrated in FIG. **8**, according to another embodiment of the present disclosure, the plurality of spoke regions **116** may each include a first spoke region **116a** extending outward from the inner region **112** in the radial direction R, a second spoke region **116b** extending from the first spoke region **116a** in the first peripheral direction P1 of the two directions of the peripheral direction P, and a third spoke region **116c** extending outward from the second spoke region **116b** in the radial direction R and connected to the outer region **114**. According to another embodiment of the present disclosure, the plurality of spoke regions **116** provided in the deformable part **110** may be understood as having an approximately S-shape or a zigzag shape. Unlike the above-mentioned embodiment of the present disclosure, a plurality of protruding sections (more specifically, two protruding sections) may be understood as being formed in each of the spoke regions **116**.

(51) More specifically, according to another embodiment of the present disclosure, as illustrated in FIG. **8**, the first spoke region **116a** may extend outward from the inner region **112** in the radial direction R and extend to be directed in the second peripheral direction P2. The third spoke region **116c** may extend outward from the second spoke region **116b** in the radial direction R and extend to be directed in the second peripheral direction P2. The second spoke region **116b** may extend approximately in parallel with the peripheral direction P.

(52) In addition, according to another embodiment of the present disclosure, a region Z23 in which the second spoke region **116b** and the third spoke region **116c** of one spoke region **116** provided in the deformable part **110** meet may be spaced apart outward, in the first peripheral direction P1 and the radial direction R, from a region Z12 in which the first spoke region **116a** and the second spoke region **116b** meet in another spoke region **116** of the two adjacent spoke regions that is spaced apart from one spoke region **116** in the first peripheral direction P1 of the peripheral direction P. In this case, as illustrated in FIG. **9**, when the spoke regions **116** are deformed during the process in which the wheel moves, the region Z23 may support the region Z12, such that the two adjacent spoke regions **116** may support each other, like the above-mentioned embodiment of the present disclosure.

(53) Hereinafter, a mobility equipped with the wheel according to embodiments of the present disclosure will be described. The above-mentioned description of the wheel according to embodiments of the present disclosure may also be equally applied to the mobility according to embodiments of the present disclosure to be described below.

(54) Mobility

(55) The mobility according to embodiments of the present disclosure may include the wheel **100**

and a frame member to which the wheel **100** is coupled.

(56) The wheel **100** may include the deformable part **110** configured to be reversibly deformed in shape by an external force.

(57) The deformable part **110** may include the inner region **112** having a shape extending in the peripheral direction P, the outer region **114** having a shape extending in the peripheral direction P and spaced apart outward from the inner region in the radial direction R, and the spoke region **116** provided between the inner region **112** and the outer region **114** and having one side connected to the inner region **112** and the other side connected to the outer region **114**.

(58) The spoke region **116** may include the first spoke region **116a** having the section protruding in the first peripheral direction P1 of the two directions of the peripheral direction P, and the second spoke region **116b** disposed at one side of the first spoke region **116a** based on the second peripheral direction P2 opposite to the first peripheral direction P1 of the two directions of the peripheral direction P and having the section protruding in the second peripheral direction P2.

(59) For example, in the case in which the wheel **100** is mounted on the mobility according to embodiments of the present disclosure, the wheel **100** may be mounted such that the mobility moves forward when the wheel **100** rotates in the second peripheral direction P2. Alternatively, the wheel **100** may be mounted such that the mobility moves forward when the wheel **100** rotates in the first peripheral direction P1.

(60) The present disclosure has been described with reference to the limited embodiments and the drawings, but the present disclosure is not limited thereto. The present disclosure may be carried out in various forms by those skilled in the art, to which the present disclosure pertains, within the technical spirit of the present disclosure and the scope equivalent to the appended claims.

Claims

1. A wheel comprising: a deformable part that can be reversibly deformed in shape by an external force, wherein the deformable part comprises: an inner region having a shape extending in a peripheral direction; an outer region having a shape extending in the peripheral direction and spaced apart outward from the inner region in a radial direction; and a spoke region provided between the inner region and the outer region and having a first side connected to the inner region and a second side connected to the outer region, wherein the spoke region comprises: a first spoke region having a section protruding in a first of two peripheral directions; and a second spoke region disposed at a first side of the first spoke region based on a second of the two peripheral directions, the second peripheral direction opposite to the first peripheral direction, wherein the second spoke region has a section protruding in the second peripheral direction, wherein a thickness of the first spoke region at a portion where the first spoke region is connected to the inner region is larger than a thickness of the first spoke region at a portion where the first spoke region is connected to the outer region.

2. The wheel of claim 1, wherein: the spoke region is one of a plurality of spoke regions, and each of the plurality of spoke regions is disposed in the peripheral direction.

3. The wheel of claim 2, wherein a first interval between the outer region and a section of the first spoke region, which protrudes by a largest amount in the first peripheral direction, is larger than a second interval between the outer region and a section of the second spoke region that protrudes by a largest amount in the second peripheral direction.

4. The wheel of claim 2, wherein a first interval between a region in which the first spoke region and the outer region meet and a region in which the second spoke region and the outer region meet is larger than a second interval between a region in which the first spoke region and the inner region meet and a region in which the second spoke region and the inner region meet.

5. The wheel of claim 2, wherein the plurality of spoke regions is disposed at intervals that allow the first spoke region, and the second spoke region spaced apart from the first spoke region in the

first peripheral direction to come into direct contact with each other when an external force of a predetermined magnitude or higher is applied to the outer region in the radial direction.

6. The wheel of claim 1, wherein the first spoke region is longer than the second spoke region.

7. The wheel of claim 1, wherein the first spoke region is thicker than the second spoke region.

8. The wheel of claim 1, wherein a first interval between the outer region and a section of the first spoke region, which protrudes by a largest amount in the first peripheral direction, is smaller than a second interval between the inner region and a section of the first spoke region that protrudes by a largest amount in the first peripheral direction.

9. The wheel of claim 1, wherein a third interval between the outer region and a section of the second spoke region, which protrudes by a largest amount in the second peripheral direction, is smaller than a fourth interval between the inner region and a section of the second spoke region that protrudes by a largest amount in the second peripheral direction.

10. The wheel of claim 1, wherein a thickness of the second spoke region at a portion where the second spoke region is connected to the inner region is larger than a thickness of the second spoke region at a portion where the second spoke region is connected to the outer region.

11. The wheel of claim 1, wherein a portion where the first spoke region is connected to the outer region is directed away in the peripheral direction from a portion where the first spoke region is connected to the inner region.

12. The wheel of claim 1, wherein a portion where the second spoke region is connected to the outer region is directed away in the peripheral direction from a portion where the second spoke region is connected to the inner region.

13. The wheel of claim 1, wherein a portion where the first spoke region is connected to the outer region is directed away in the second peripheral direction from a portion where the first spoke region is connected to the inner region.

14. The wheel of claim 1, wherein a portion where the second spoke region is connected to the outer region is directed away in the second peripheral direction from a portion where the second spoke region is connected to the inner region.

15. The wheel of claim 1, further comprising: a core part provided inside the inner region in the radial direction and having a first side fixedly coupled to the inner region; and a connection part having a first side connected to the core part and a second side connected to the outer region.

16. A wheel comprising: a deformable part that can be reversibly deformed in shape by an external force, wherein the deformable part comprises: an inner region having a shape extending in a peripheral direction; an outer region having a shape extending in the peripheral direction and spaced apart outward from the inner region in a radial direction; and spoke regions provided between the inner region and the outer region, each of the spoke regions having a first side connected to the inner region and a second side connected to the outer region, wherein the spoke regions are spaced apart from one another in the peripheral direction, and wherein each of the spoke regions comprises: a first spoke portion extending outward from the inner region in the radial direction; a second spoke portion extending from the first spoke portion in a first peripheral direction of two directions of the peripheral direction; and a third spoke portion extending outward from the second spoke portion in the radial direction and connected to the outer region.

17. The wheel of claim 16, wherein: the first spoke portion extends outward from the inner region in the radial direction and extends to be directed in a second peripheral direction opposite to the first peripheral direction of the peripheral direction; and the third spoke portion extends outward from the second spoke portion in the radial direction and extends to be directed in the second peripheral direction.

18. The wheel of claim 16, wherein a first region in which the second spoke portion and the third spoke portion of a first spoke region of the spoke regions meet is spaced apart outward, in the radial direction, from a second region in which the first spoke portion and the second spoke portion

meet in a second spoke region of the spoke regions that is spaced apart from the first spoke region in the first peripheral direction of the peripheral direction.
